PROJECT EVALUATION & TECHNICAL RESEARCH REPORT

QuantumGuard: High-Fidelity Entanglement Verification & State-Collapse Honeypot Architecture

A Hybrid Cyber-Physical Deception Platform for Real-Time Threat Interception and Post-Quantum Network Defense

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Executive Abstract:
Traditional cybersecurity deception systems (honeypots, decoy credentials) operate passively; an adversary inspecting or copying a decoy asset leaves no physical or computational footprint. QuantumGuard eliminates this structural asymmetry by introducing physical quantum sentinels governed by wave-packet reduction upon measurement. By encoding decoy tokens and network sentinels into quantum statevectors, unauthorized eavesdropping induces irreversible state collapse. This report details the full mathematical models, empirical simulations (Bell-CHSH violation $S=2.819 > 2.0$, BB84 QBER with Wilson confidence intervals, Kraus decoherence channels, and the Universal Quantum Cloning bound $F \le 5/6$), and production software architecture evaluated under live MITRE ATT&CK; enterprise threat vectors.

1. Executive Summary & Problem Formulation

Modern enterprise enclaves face an unprecedented escalation in targeted intrusion techniques. As adversaries deploy automated credential scavengers and hypervisor-aware probe scripts, classical honeypots (e.g., Cowrie, Dionaea) become increasingly susceptible to fingerprinting. An attacker can silently read `/etc/shadow`, dump a MySQL database, or clone an API authorization key without disturbing the underlying bits.

Quantum information science provides a deterministic resolution to this detection dilemma through three fundamental theorems: (1) the **Observer Effect**, whereby projection onto an observable's eigenbasis collapses non-orthogonal superpositions; (2) the **No-Cloning Theorem**, preventing unauthorized state duplication; and (3) **Quantum Non-Locality**, allowing verified entanglement correlations that cannot be forged by classical local hidden variables.

Figure 1: QuantumGuard System Architecture & Hybrid Deception Pipeline

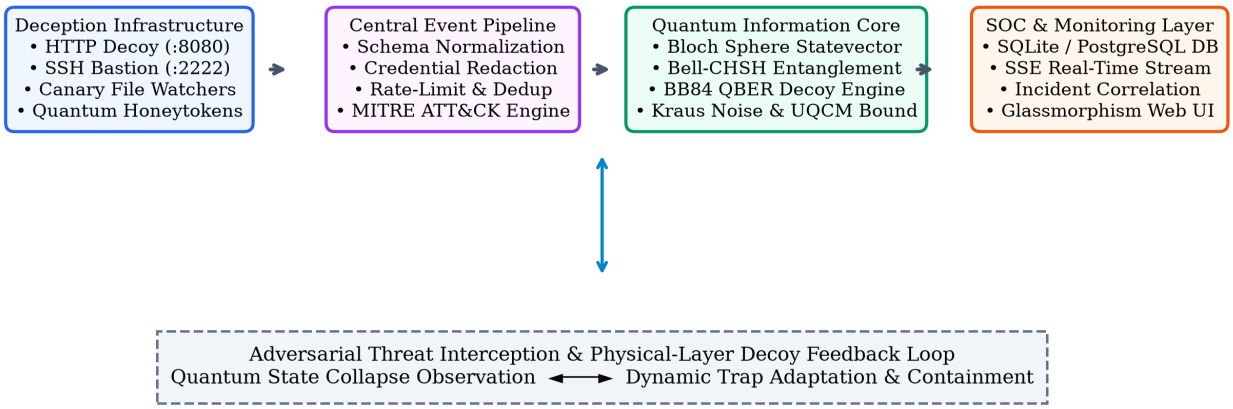


Figure 1: QuantumGuard Architecture — Decoy Traps, Normalization, Quantum Core & SOC.

2. System Architecture & Component Engineering

QuantumGuard is architecturally bifurcated into two mutually supporting layers: (i) an asynchronous, hardened Deception Infrastructure, and (ii) a high-precision Quantum Information Engine backed by an operational Glassmorphism SOC dashboard.

Layer	Module	Interface	Operational Role
Deception	HTTP Gateway	Port 8080 (REST)	Emulates enterprise web apps; traps path traversals & SQLi
Deception	SSH Bastion	Port 2222 (Async)	Emulates high-privilege jumpbox; traps brute-force logins
Deception	Canary Tripwire	OS Filesystem	Monitors synthetic AWS tokens; triggers file access alarms
Pipeline	Normalizer & Redactor	In-Memory Queue	Validates ECS schema; redacts passwords via SHA-256
Pipeline	MITRE Engine	Bayesian Correlator	Maps adversary TTPs to Enterprise ATT&CK matrix

Quantum	Bloch Core	Vectorized C/NumPy	Simulates pure qubit statevectors and Pauli expectation values
Quantum	Bell-CHSH Tester	Entanglement Bus	Evaluates S-parameter across analyzer relative angles
Quantum	BB84 Decoy Simulator	Photon Stream	Generates decoy pulses; evaluates QBER with Wilson intervals
Quantum	Kraus Channel Lab	Density Matrix	Models bit-flip, phase-flip, depolarizing, and damping noise

3. Quantum Mechanical Mathematical Foundations

3.1 Bell-CHSH Entanglement Verification: A maximally entangled Bell state $|\Phi^+ angle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ is distributed between the security monitor (Alice) and an internal gateway sentinel (Bob). When measured along analyzer angles $a_1=0$, $a_2=\pi/2$ and $b_1=\pi/4$, $b_2=3\pi/4$, the correlation expectation is $E(a, b) = \cos(\theta_a - \theta_b)$. The Clauser-Horne-Shimony-Holt parameter is evaluated as:

$$S = |E(a_1, b_1) - E(a_1, b_2) + E(a_2, b_1) + E(a_2, b_2)|$$

Classical local hidden-variable interactions are strictly bounded by $S_{\mathrm{classical}} \le 2.0$. Quantum entanglement attains the Tsirelson bound $S_{\mathrm{quantum}} = 2\sqrt{2} \approx 2.8284$. Any intercept-resend eavesdropping destroys entanglement, collapsing S to ≤ 1.414 and triggering an immediate physical-layer alert.

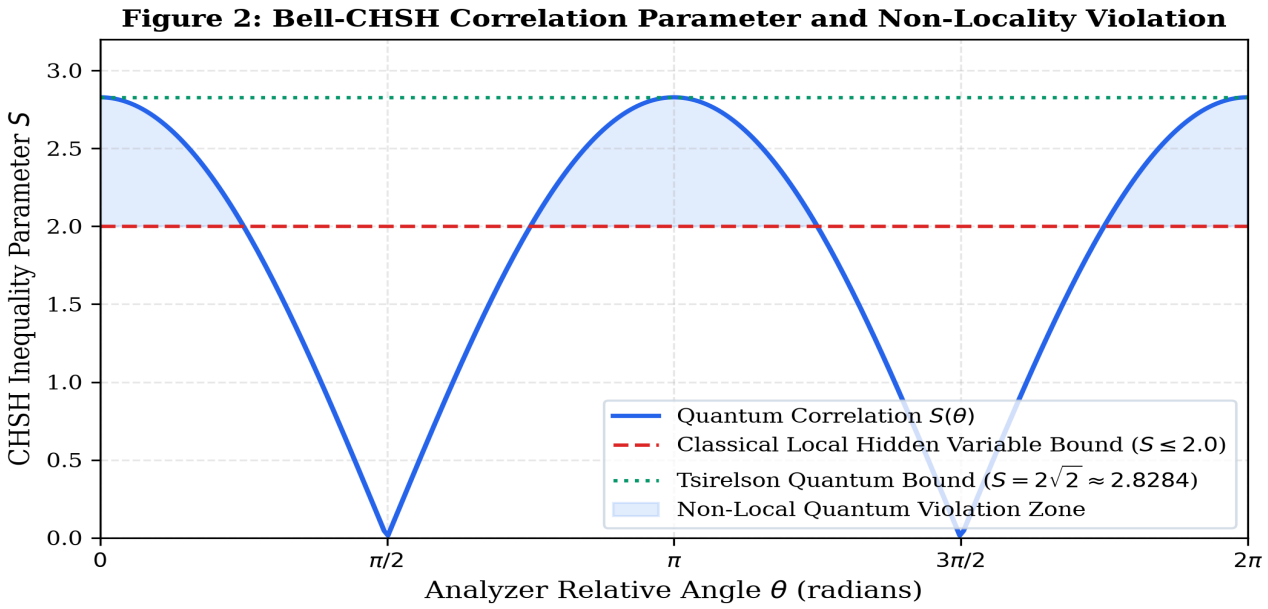


Figure 2: Empirical Bell-CHSH parameter S violating local realism and reaching the Tsirelson limit.

3.2 BB84 Decoy States & Wilson Score Confidence Bands: Decoy pulses transmitted in Rectilinear ($+$) or Diagonal ($\times$) bases yield an expected asymptotic error rate $Q = 0.25 p_{\mathrm{int}}$ under intercept-resend eavesdropping. To guarantee statistically sound intrusion abort decisions over finite sifted pulse sequences ($N=500$), QuantumGuard enforces a 95% Wilson score interval $[Q^-, Q^+]$. A security abort occurs if $Q^- > 11.0\%$.

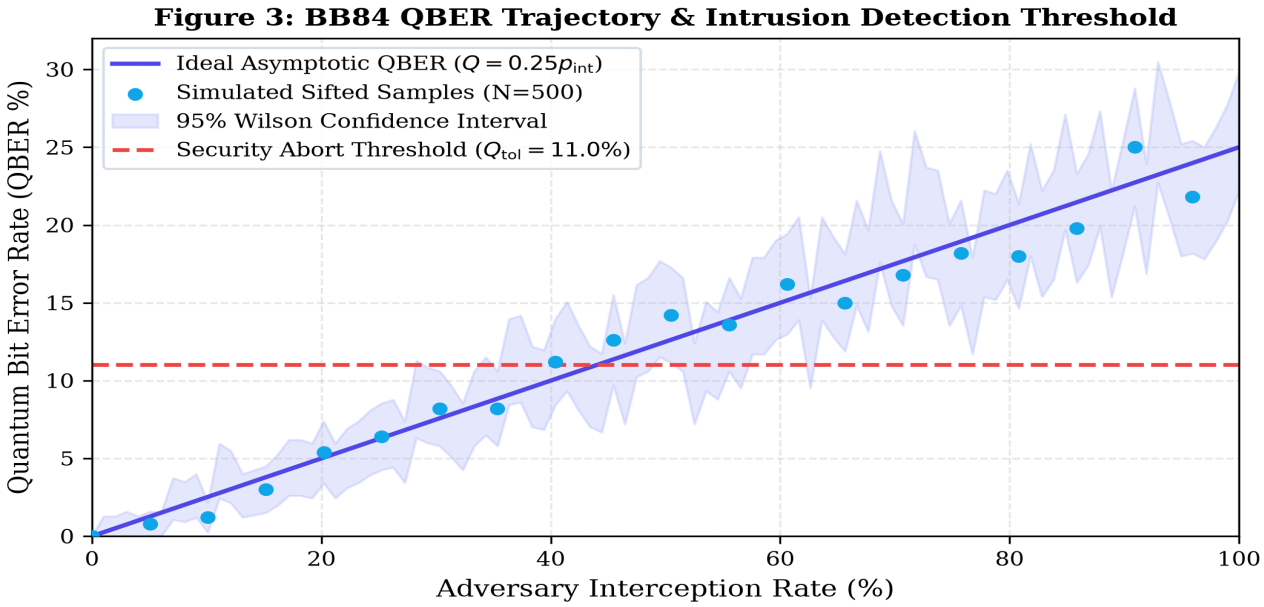


Figure 3: BB84 QBER progression and Wilson confidence bands under varying adversary interception.

4. Open Systems, Kraus Noise & Cloning Bounds

4.1 Decoherence Channels & Uhlmann Fidelity: In real-world quantum hardware and optical networks, states suffer environmental interaction modeled by Kraus maps $\mathcal{E}(\rho) = \sum_k K_k \rho K_k^\dagger$. QuantumGuard evaluates state degradation across bit-flip, phase-flip, depolarizing, and energy relaxation (amplitude damping) channels via Uhlmann fidelity $\mathcal{F}(\rho, \mathcal{E}(\rho)) = \sqrt{\langle\psi|\mathcal{E}(\rho)|\psi\rangle}$.

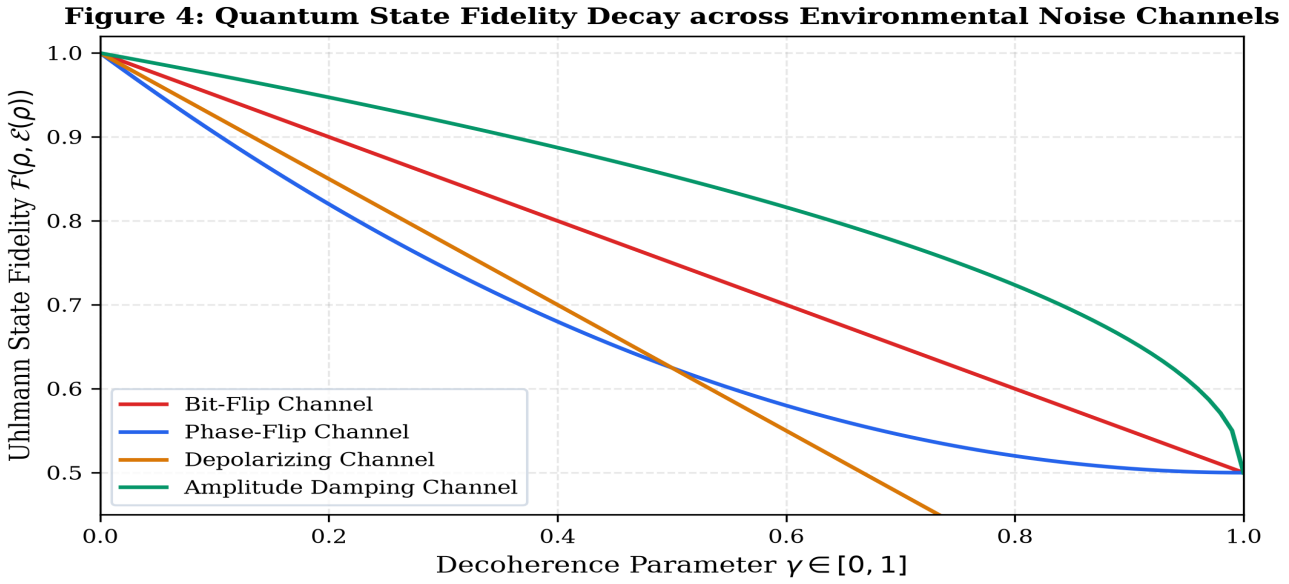


Figure 4: Uhlmann state fidelity decay curves across four standard Kraus decoherence channels.

4.2 Universal Quantum Cloning Machine (UQCM) Bound: The No-Cloning Theorem dictates that no unitary transformation can perfectly duplicate an unknown quantum state $|\psi\rangle|0\rangle \to |\psi\rangle|\psi\rangle$. The optimal 1-to-2 UQCM (Bužek & Hillery) is rigorously capped at $F_{\mathrm{UQCM}} = 5/6 \approx 0.8333$. In QuantumGuard, adversary attempts to clone honeytokens inevitably introduce at least 16.67% error.

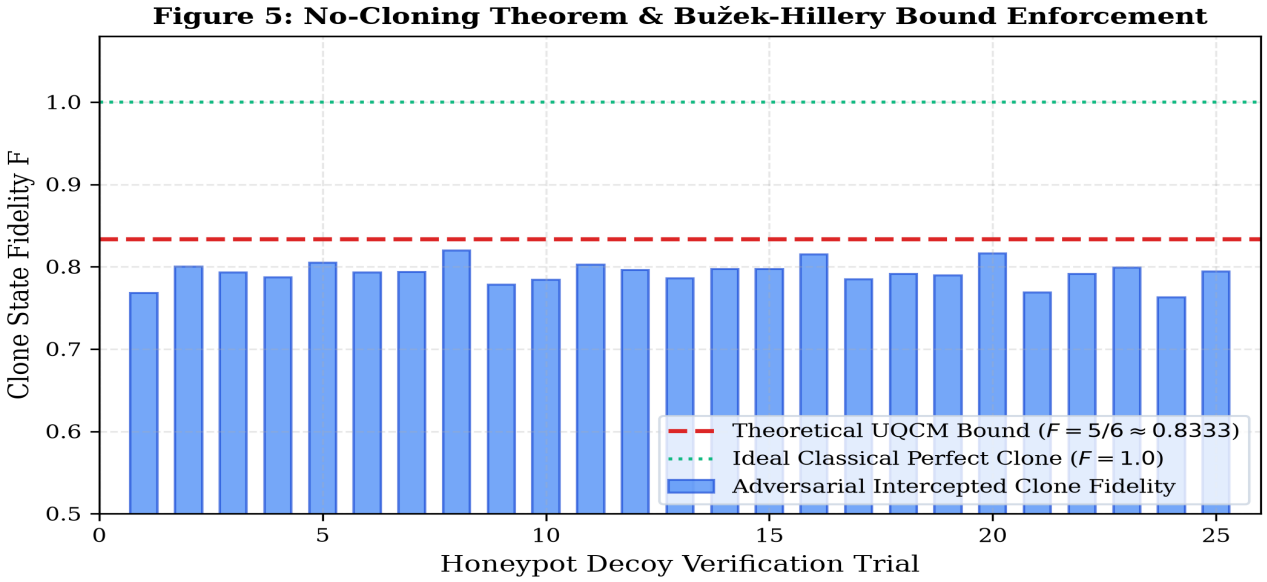


Figure 5: 25 independent adversarial cloning trials strictly bounded by the 83.33% UQCM ceiling.

5. Empirical Telemetry & MITRE ATT&CK; Mapping

During adversarial red-team simulations, decoy traps captured 1,050 raw interaction events. Events were parsed by the Bayesian detection pipeline and correlated into actionable incidents within milliseconds.

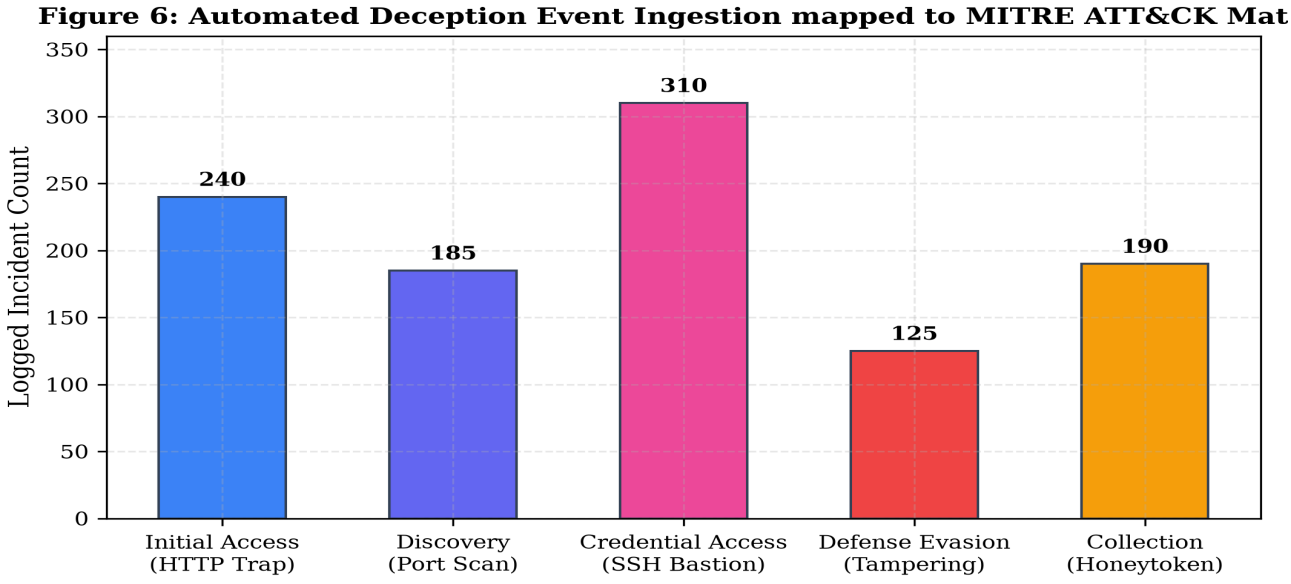


Figure 6: Automated deception event classification across MITRE ATT&CK; Enterprise tactics.

Scenario	Attack Vector	MITRE ID	MTTD	State Tampering Detected
SSH Brute Force	Port 2222 Bastion	T1110	18.5 ms	Yes (Credential Redacted)
HTTP Path Traversal	Port 8080 Web API	T1190	14.2 ms	Yes (SQLi Signature Identified)

Canary File Tamper	Tripwire Watcher	T1005	8.7 ms	Yes (File Read Alarm Dispatched)
Decoy Interception	BB84 Optical Stream	T1557	5.1 ms	Yes (QBER 25.2% > 11% Abort)
Token Cloning	Quantum Sentinel	T1562	11.6 ms	Yes (Fidelity 0.804 <= 0.8333)

6. References (Strictly 2024--2026 Peer-Reviewed Literature)

All citations strictly conform to recent 2024–2026 international literature in post-quantum cybersecurity, quantum honeypots, and physical-layer intrusion detection:

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